

PROBLEMA 2

$$1) S \rightarrow 0A0 \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

a) eliminar epsilon

$$S \rightarrow 00 \mid 0A0 \mid 11 \mid 1B1 \mid B \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

b) eliminar unarias

$$(S, S) \quad S \rightarrow B \quad (S, B) \quad (B, B) \quad B \rightarrow A \quad (B, A)$$

$$(S, B) \quad B \rightarrow A \quad (S, A) \quad (B, A) \quad A \rightarrow C \quad (B, C)$$

$$(S, A) \quad A \rightarrow C \quad (S, C) \quad (B, C) \quad C \rightarrow S \quad (B, S)$$

$$(S, C) \quad C \rightarrow S \quad (S, S) \quad (B, S) \quad S \rightarrow B \quad (B, B)$$

$$(A, A) \quad A \rightarrow C \quad (A, C) \quad (C, C) \quad C \rightarrow S \quad (C, S)$$

$$(A, C) \quad C \rightarrow S \quad (A, S) \quad (C, S) \quad S \rightarrow B \quad (C, B)$$

$$(A, S) \quad S \rightarrow B \quad (A, B) \quad (C, B) \quad B \rightarrow A \quad (C, A)$$

$$(A, B) \quad B \rightarrow A \quad (A, A) \quad (C, A) \quad A \rightarrow C \quad (C, C)$$

$$S \rightarrow 00 \mid 0A0 \mid 11 \mid 1B1 \mid BB$$

$$A \rightarrow 00 \mid 0A0 \mid 11 \mid 1B1 \mid BB$$

$$B \rightarrow 00 \mid 0A0 \mid 11 \mid 1B1 \mid BB$$

$$C \rightarrow 00 \mid 0A0 \mid 11 \mid 1B1 \mid BB$$

c) eliminar inútiles e inutilizables

$$T = \{0, 1\}$$

$$W_1 = \{B, A, B, C\}$$

$$W_2 = \{B, A, B, C\}$$

$$S \rightarrow 0010A01111B1BB$$

$$A \rightarrow 0010A01111B1BB$$

$$B \rightarrow 0010A01111B1BB$$

d) CNF

$$C \rightarrow 0$$

$$C_1 \rightarrow AC$$

$$S \rightarrow CC | CC_1 | DD | DC_2 | BB$$

$$A \rightarrow CC | CC_1 | DD | DC_2 | BB$$

$$D \rightarrow 1$$

$$C_2 \rightarrow BD$$

$$B \rightarrow CC | CC_1 | DD | DC_2 | BB$$

2) $S \rightarrow aAa | bBb | \epsilon$
 $A \rightarrow C | a$
 $B \rightarrow C | b$
 $C \rightarrow CDE | \epsilon$
 $D \rightarrow A | B | ab$

a) eliminación epsilon

$S \rightarrow aa | aAa | bb | bBb$
 $A \rightarrow C | a$
 $B \rightarrow C | b$
 $C \rightarrow CDE | DE | CE | E$
 $D \rightarrow A | B | ab$

b) uniones

$(A,A) \quad A \rightarrow C \quad (A,C)$	$(C,C) \quad C \rightarrow E \quad (C,E)$
$(A,C) \quad C \rightarrow E \quad (A,E)$	
	$(D,D) \quad D \rightarrow A \quad (D,A)$
$(B,B) \quad B \rightarrow C \quad (B,C)$	$(D,A) \quad A \rightarrow C \quad (D,C)$
$(B,C) \quad C \rightarrow E \quad (B,E)$	$(D,C) \quad C \rightarrow E \quad (D,E)$
	$(D,D) \quad D \rightarrow B \quad (D,B)$
	$(D,B) \quad B \rightarrow C \quad (D,C)$

$S \rightarrow aa | aAa | bb | bBb$
 $A \rightarrow CDE | DE | CE | a$
 $B \rightarrow CDE | DE | CE | b$
 $C \rightarrow CDE | DE | CE$
 $D \rightarrow a | b | CDE | DE | CE | ab$

c) No produce / incompatible

$$T = \{a, b\}$$

$$W_1 = \{S, A, B, D\}$$

$$W_2 = \{S, A, B, D\}$$

$$Y = \{a\}$$

$$Y_1 = \{S, A, B, a, b\}$$

$$Y_2 = \{S, A, B, a, b\}$$

$$S \rightarrow aa|aAa|bb|bBb$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$D \rightarrow a|b|ab$$

$$S \rightarrow aa|aAa|bb|bBb$$

$$A \rightarrow a$$

$$B \rightarrow b$$

d) CNF

$$C_1 \rightarrow AA$$

$$C_2 \rightarrow BB$$

$$S \rightarrow AA|AC_1|BB|BC_2$$

$$A \rightarrow a$$

$$B \rightarrow b$$

3) $S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S$
 $B \rightarrow b \mid \epsilon$

q) epson

$$\begin{aligned} S &\rightarrow SA \mid AS \mid S \mid ASA \mid a \mid aB \\ A &\rightarrow BIS \\ B &\rightarrow b \end{aligned}$$

b) Unenig

(S, S)	$S \rightarrow S$	(S, S)	$S \rightarrow SA AS ASA a aB$
			$A \rightarrow SA AS ASA a aB b$
(A, A)	$A \rightarrow B$	(A, B)	$B \rightarrow b$
(A, A)	$A \rightarrow S$	(A, S)	

e) Nepeanuren / Inalenzzeiten

d) eNF

$$\begin{array}{ll} T = \{a, b\} & Y = \{S\} \\ W_1 = \{S, A, B\} & Y_1 = \{S, A, B', a\} \\ W_2 = \{S, A, B\} & Y_2 = \{S, A, B, a, b\} \\ & Y_3 = \{S, A, B, a, b\} \end{array}$$
$$\begin{aligned} C &\rightarrow a \\ C_1 &\rightarrow SA \\ B &\rightarrow SA | AS | Ae_1 | a | CB \\ A &\rightarrow SA | AS | Ac_1 | a | CB | b \\ B &\rightarrow b \end{aligned}$$